Industrial Visual Inspection: From Prototype to Operational Decision

Executive summary · independent project on public benchmark images (VisA), not factory data

Engineering milestones are dated from the development repository; the project owner's review of this work is pending.

The problem

Circuit-board defects are rare and varied, so there is no large labelled defect library. The question explored: can a model that has only seen good boards send the right images to a human reviewer without burying the reviewer in false alarms?

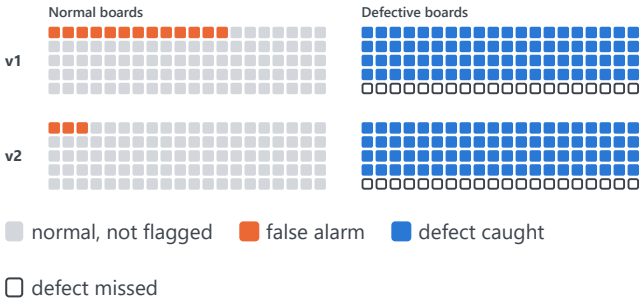
What was built

A working prototype: frozen pretrained CNN features, a per-position Gaussian (Mahalanobis) model of normal appearance, board registration and — in v2 — a score taken over the board region only. Thresholds are frozen on separate normal images before testing; every change was pre-registered and evaluated once on a board type not used in development.

Measured result — held-back board type PCB2

100 defective + 100 normal test images; 95% confidence intervals

	v1	v2	Change
Defect recall	80%	**80%**	unchanged
False-alarm rate (test normals)	13%	**3%**	−10 pts (CI −17 to −3)
False-alarm rate (holdout normals)	5.3%	**4.0%**	−1.3 pts (CI −4.6 to +2.0)
AUROC	0.927	**0.979**	+0.052 (CI +0.028 to +0.082)



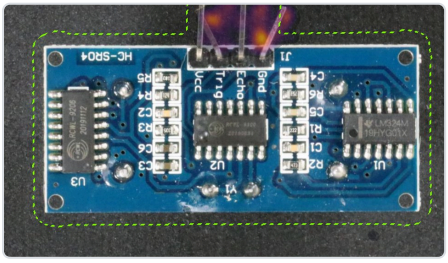
Median end-to-end latency per image (GPU, one image at a time): v1 194 ms, v2 224 ms. Although v2 improved false-positive performance, it did not meet the proposed detection-sensitivity and robustness requirements. It is therefore not recommended for a manufacturing pilot without further work. Recall stays below the proposed 90% target; rule-based verdict against the unchanged criteria: INCONCLUSIVE.

Remaining technical risks

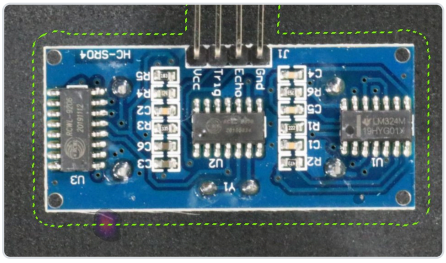
- **Detection.** Recall 80% against a proposed 90%; the smallest third of defects only 56%.
- **Lighting.** A 10% darker image raised PCB2 false alarms from 3.6% to 28.7%.
- **Portability.** On PCB3's green background the original board-finding rule marked 96% of the frame as board, silently.
- **Board region.** About 31% of the PCB3 scoring region is background cloth; lint near the board can still raise a false alarm.

RECOMMENDATION

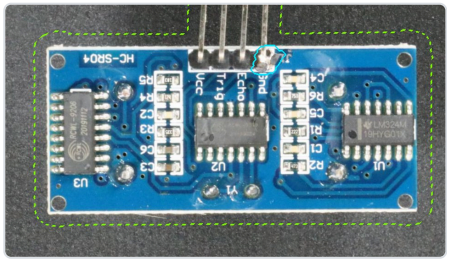
No manufacturing pilot recommended yet. Go for one bounded validation step: agree requirements and costs with a quality engineer, then run the tool in shadow mode on real line images, with the site checks on the next page as entry gates. Decide again on that evidence.



Defect caught: score 88.6 vs threshold 51.2



False alarm (peak on cloth near board): score 65.1 vs threshold 51.2



Small defect missed (cyan = ground truth): score 31.4 vs threshold 51.2

Follow-up experiment: lighting correction on a new board type (PCB3)

One pre-registered change — standardise each image's global brightness and contrast before registration — compared with the unchanged pipeline on 256 unseen normal and 100 defective images, evaluated once.

Condition	False alarms, baseline	False alarms, corrected	Role
Clean	4.3%	**3.5%**	reference; recall baseline 90/100, corrected 91/100
Global darkening (G1)	8.6%	**3.1%**	primary endpoint
Brightening (B1)	19.5%	**6.2%**	secondary
Non-uniform lighting (N1)	6.2%	**5.9%**	secondary, out of class
Mild blur	4.7%	**18.0%**	acceptance check only

- **A. Pre-registered outcome:** "ROBUSTNESS IMPROVEMENT SUPPORTED" (darkening difference −4.7 pts, CI −8.6 to −0.8) — **weakly informative**: the uncorrected pipeline barely failed on PCB3, and a global darkening is exactly what a global correction undoes.
- **B. Engineering decision: do not adopt the correction.** It made a mild blur raise false alarms by +14.5 pts (uncorrected +0.4), beyond the original 10-pt limit. Non-uniform lighting was not improved; part of the effect is likely more stable registration.
- **The baseline is not production-ready either:** brightening raised its false alarms by +15.2 pts; both variants are INCONCLUSIVE against the unchanged criteria.
- **Not a progression:** PCB1, PCB2 and PCB3 are separate board types with their own models and thresholds. The PCB3 baseline's 90% recall is not comparable with PCB2's 80% and does not predict recall on a real line.
- **Disclosure:** after PCB3 test scoring, one input check in the analysis program was repaired (7dbd302); no score, threshold or decision changed, and an independent recomputation agrees on 90/90 checks (repository: docs/pcb3_post_evaluation_repair.md).

What a pilot site must prove first

- Board segmentation and registration on the site's own normal images (background, fixture, orientation).
- A lighting specification plus a per-image brightness/contrast check, and a perturbation test at the site's frozen threshold.
- A labelled sample of the site's real defects, with recall reported by defect size.
- Agreed cost of a miss versus a review, review capacity, and versioned model + threshold releases with rollback.

Delivery lifecycle — status

1. Preliminary value and Go/No-Go	Demonstrated
2. Evidence-based understanding of user needs	Proposed
3. First-Time-Right alignment	Proposed
4. Model selection and implementation	Demonstrated
5. Controlled validation	Demonstrated
6. Value realization and Go/No-Go	Pending
7. Cross-site readiness, maintenance and handoff	Proposed

Not yet validated: Any factory image, camera, lamp, line speed or real defect mix. · Usability of the review interface; reviewer agreement. · Labour saving or financial return. · Robustness to real (not synthetic) lighting changes.

"ROI" here means the image region of interest. No financial return on investment or labour saving is claimed: only a classification-cost break-even range was computed (PCB2 v2 beats reviewing nothing or everything when k = defect odds × miss cost ÷ review cost lies between 0.0375 and 4.85).

Data: VisA dataset, Zou et al., ECCV 2022, CC BY 4.0. Benchmark board types PCB1–PCB3 are not manufacturing sites. Not affiliated with or endorsed by the dataset authors or any company.

Built with AI coding assistance (Claude Code) under the project owner's direction; owner review pending. Project owner: Leon Lai.